



DevOps

DevOps Terminology

History of DevOps

- The idea began in **2008** with a discussion between **Patrick Debois** and **Andrew Clay Shafer** concerning the concept of agile infrastructure. However, the idea only started to spread in **2009** with the advent of the first **DevOpsDays** event held in Belgium.
- The revolutionary aspect of DevOps is that it crosses a traditional line by merging **software development** with the environment in which it is developed. In fact, it is the automation that makes the **agile development cycle** possible by merging different functions from software testing to deployment.



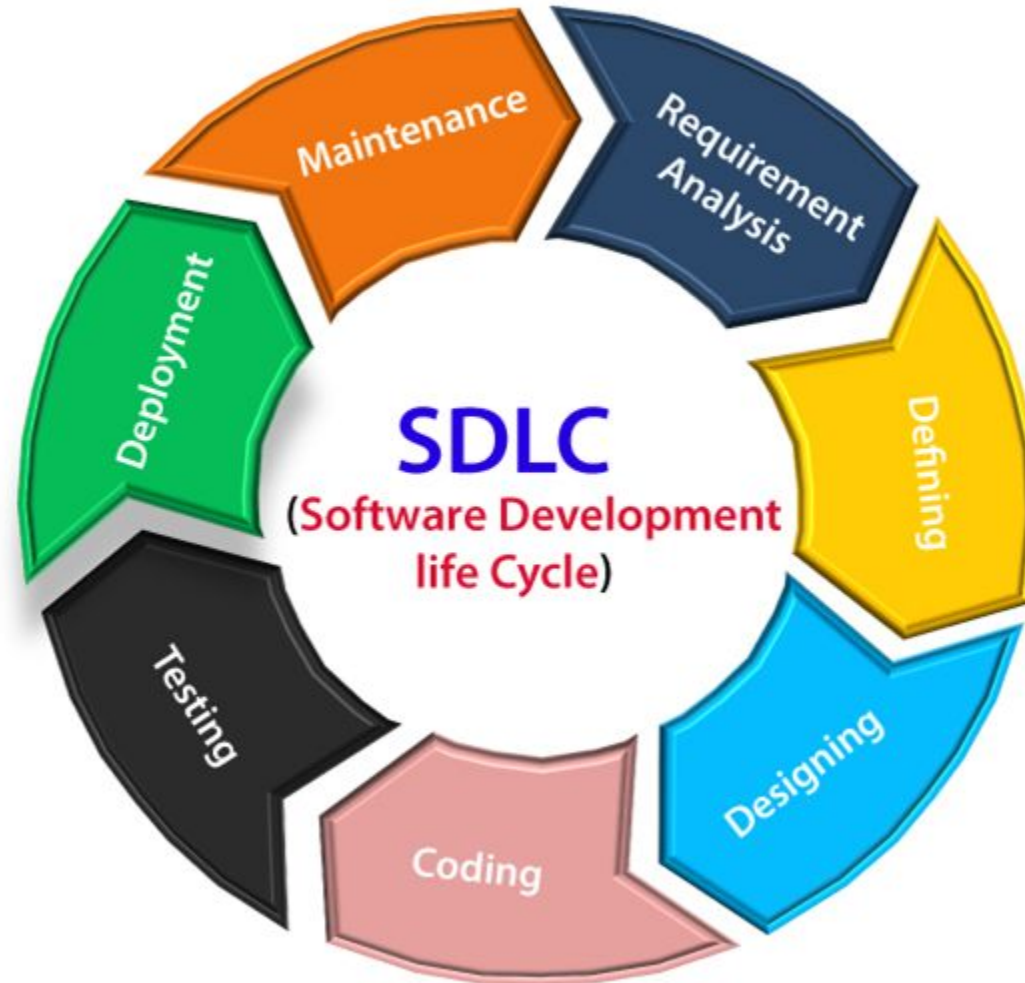
Prerequisite to learn DevOps

- To learn DevOps there are some requirements of knowledge that you need to have:
 - Git (version control)
 - Basic Concept of OS
 - Fundamental of Linux
 - Programming or scripting knowledge
 - Server Management
 - Project Lifecycle
 -

What and Why DevOps

- **DevOps** come from **Development + Operations**. It is not tools, programming language or software, but it is the idea of working to solve the problems between ***developers and operations team***. This idea officially implementing by many tools.
- DevOps integrates developer and operation team in order to improve **collaboration** and **productivity** by automating infrastructure, automating workflow and continuously measuring the application performance.

Software Development Life Cycle



Software Development Life Cycle

- **SDLC** stands for **Software Development Life Cycle**, which is a process followed by software development organizations to develop high-quality software. It is a **series of steps** that involve **planning, designing, building, testing**, and **maintaining** software.

The goal of SDLC is to ensure that software is **delivered on time, within budget**, and meets the required specifications and **quality standards**.

Software Development Life Cycle

SDLC consists of several stages, including:

- **Requirements gathering and analysis:** In this stage, the software requirements are gathered and analyzed to ensure that the software development process is properly scoped and aligned with the needs of the business.
- **Design:** In this stage, the software architecture and design are created, including the user interface and database design.
- **Development:** In this stage, the software is actually built, including **coding** and **testing** of individual modules.

Software Development Life Cycle

- **Testing:** In this stage, the software is tested to ensure that it meets the specified requirements and works as intended. This may include unit testing, integration testing, system testing, and user acceptance testing.
- **Deployment:** In this stage, the software is deployed to the production environment, where it is made available to end-users.
- **Maintenance:** In this stage, the software is monitored and maintained to ensure its continued functionality and to address any issues that may arise.

Software Development Methodology

- **Software Development Methodology**

refers to the systematic and organized approach followed in the development of software.

It provides a roadmap for the software development process, outlining the steps and activities involved in delivering a high-quality software product.



Software Development Methodology

There are several software development methodologies, including:

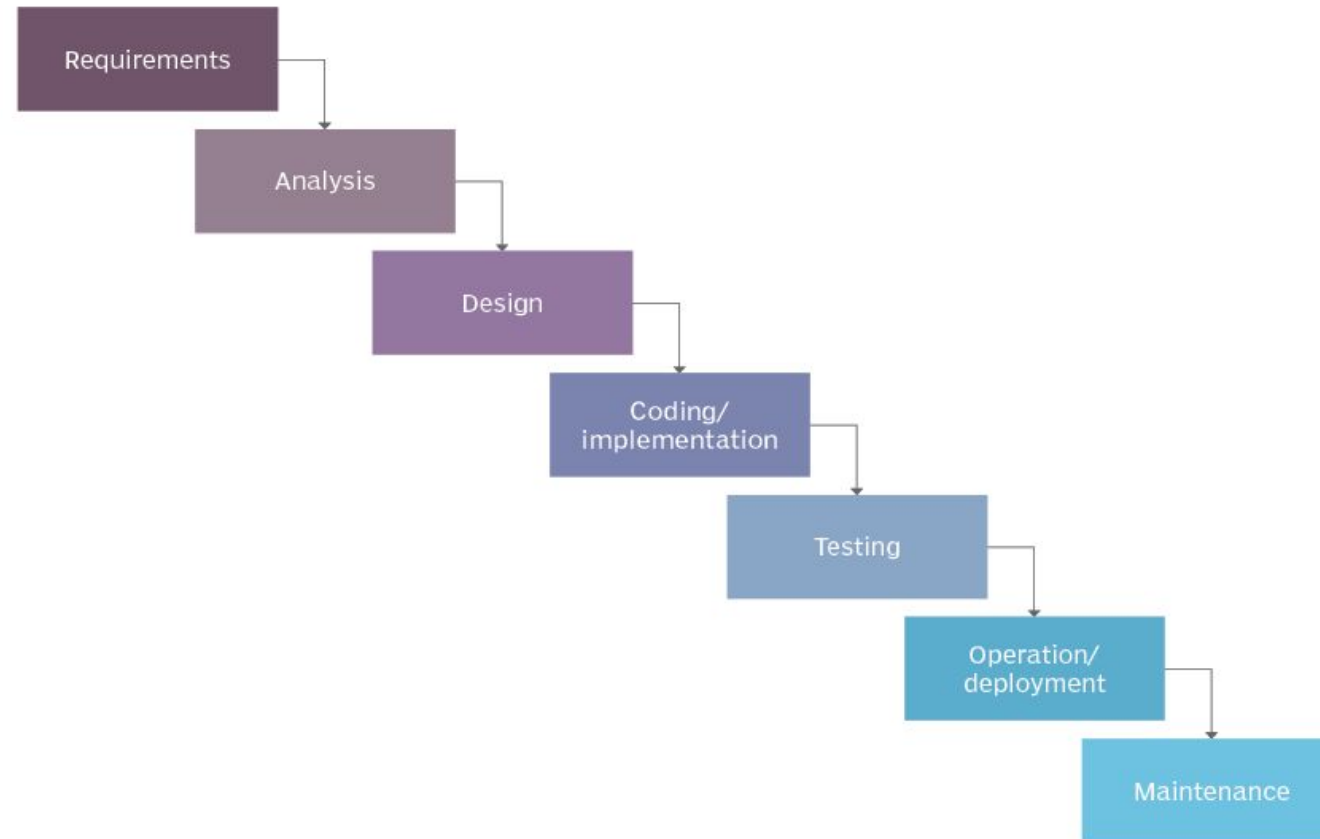
- **Waterfall:** This is a traditional, sequential approach to software development that involves a linear progression from one phase of development to the next, without revisiting earlier phases.
- **Agile:** This is a flexible and iterative approach to software development that emphasizes collaboration, customer satisfaction, and adaptability to change. It involves regularly reassessing and adapting to changing requirements and customer feedback.

Software Development Methodology

- **Hybrid:** This is a combination of different methodologies, tailored to the specific needs and constraints of a project. It may involve elements of Waterfall and Agile, or a combination of other methodologies.
- **DevOps:** This is a set of practices and processes that aim to automate and optimize the software development and deployment process, enabling faster and more reliable delivery of software.

Software Development Methodology

Waterfall model

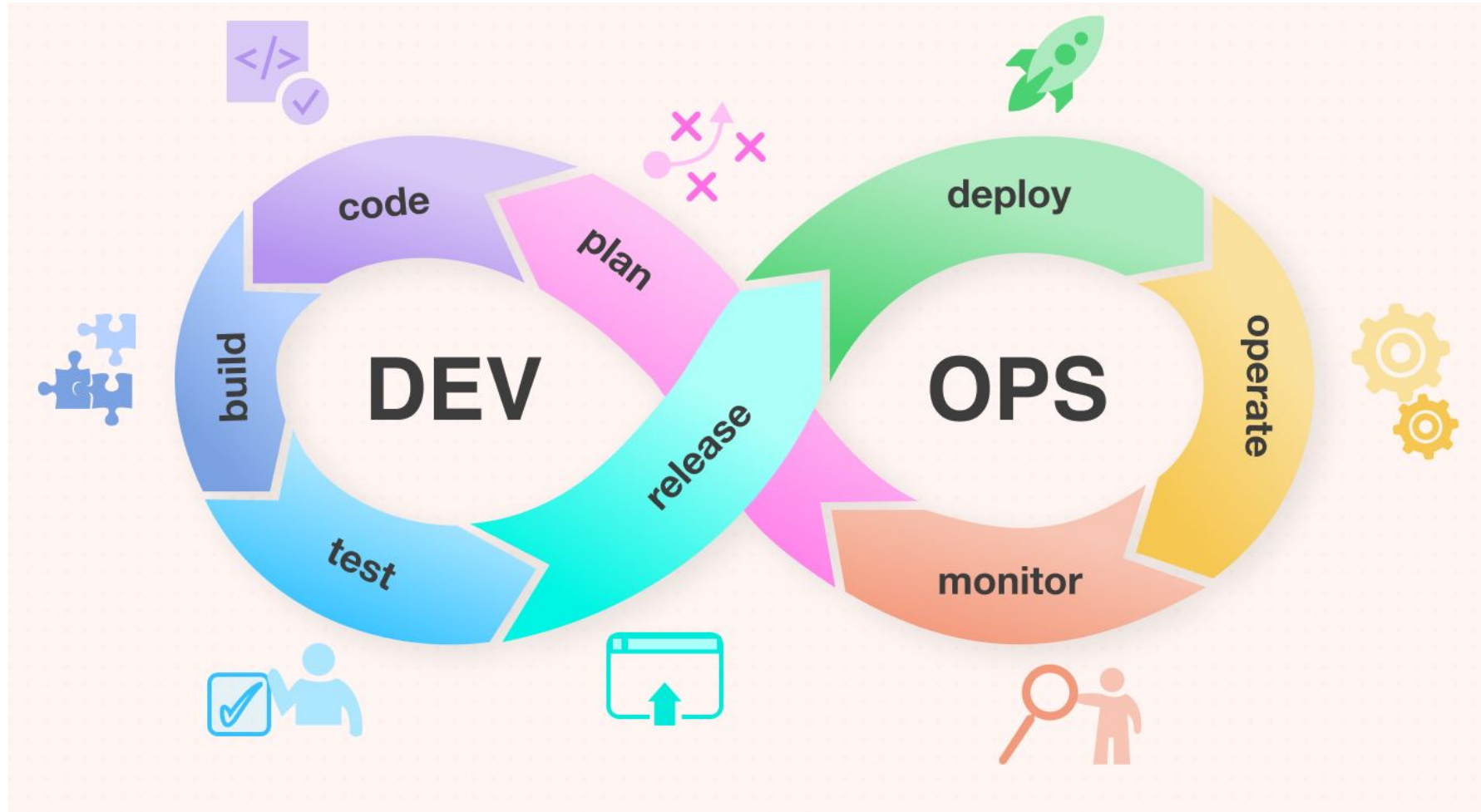


Software Development Methodology

Agile Methodology



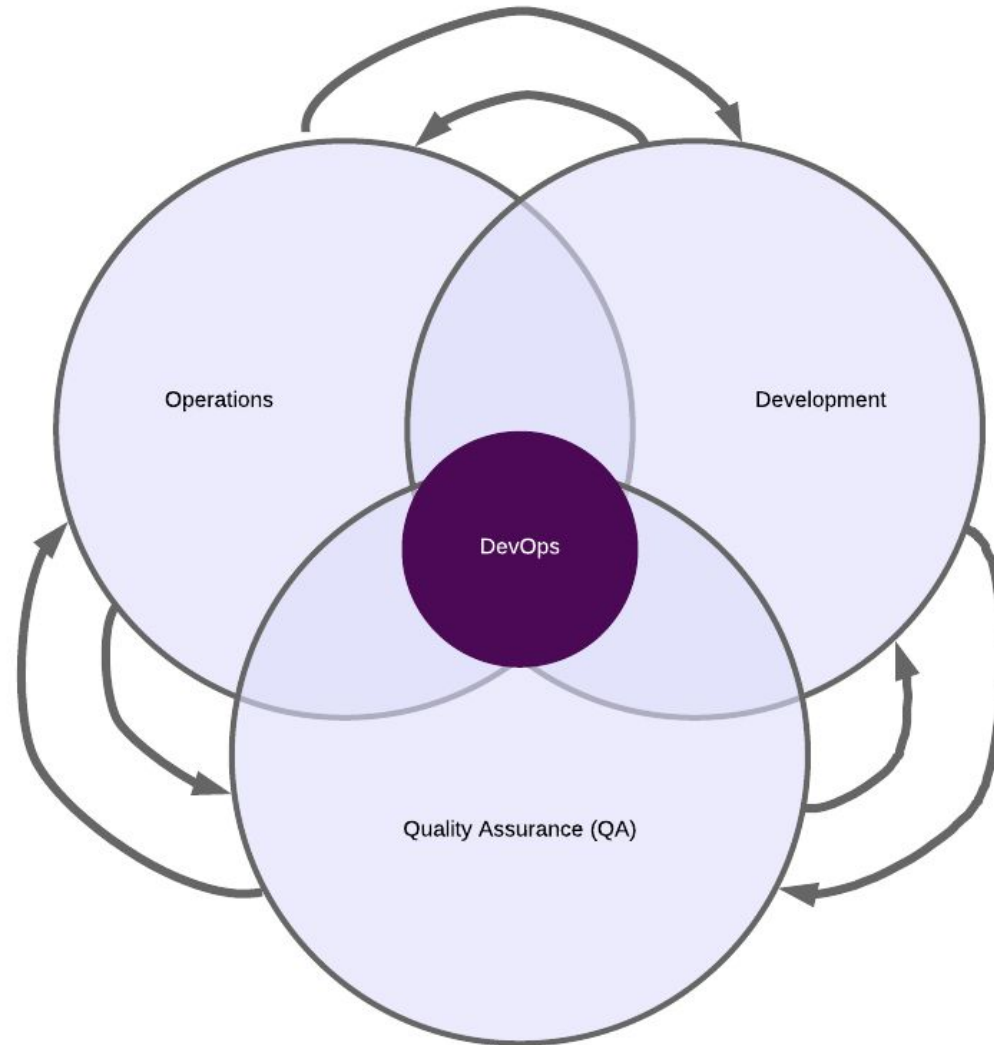
Software Development Methodology



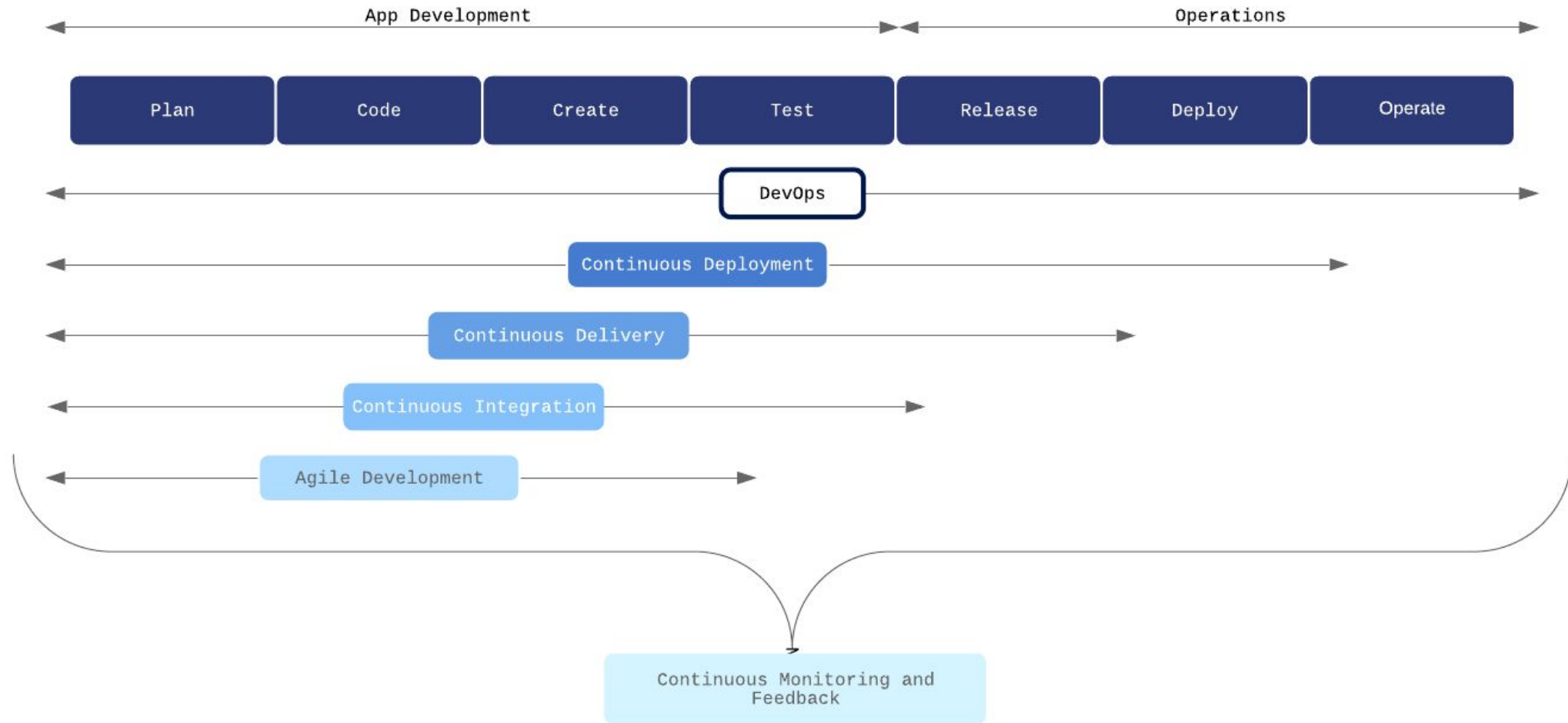
Agile Vs DevOps

While Agile *focuses primarily on the software development process*, DevOps extends that *focus to include the entire software delivery pipeline*, including testing, deployment, and operations. The goal of DevOps is to **automate** and **optimize** the software delivery process, enabling faster and more reliable delivery of high-quality software.

DevOps Methodology

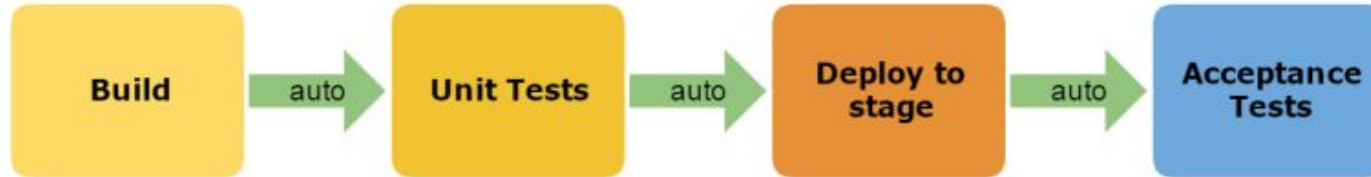


DevOps Methodology



CI & CD

Continuous Integration



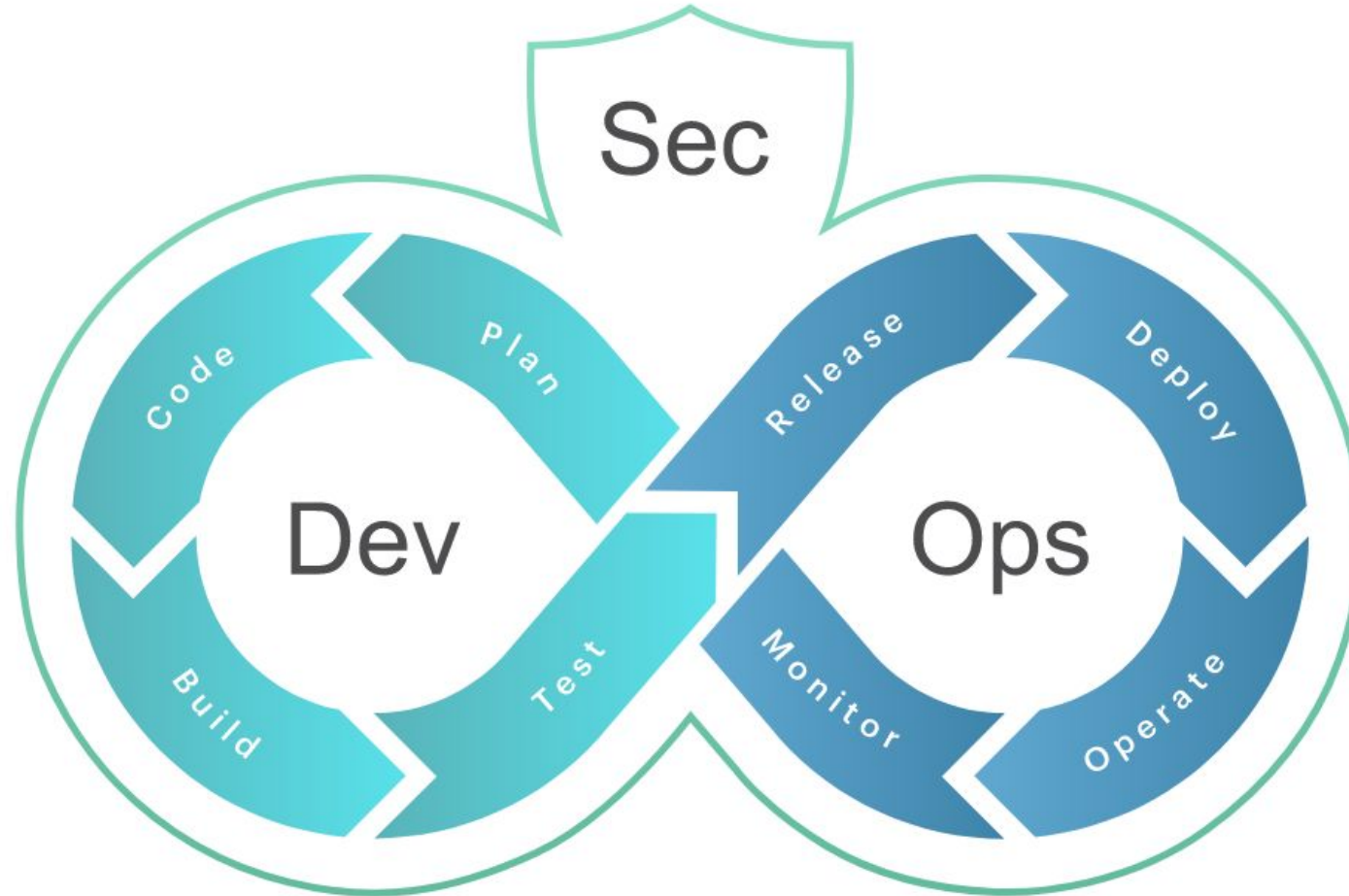
Continuous Delivery



Continuous Deployment

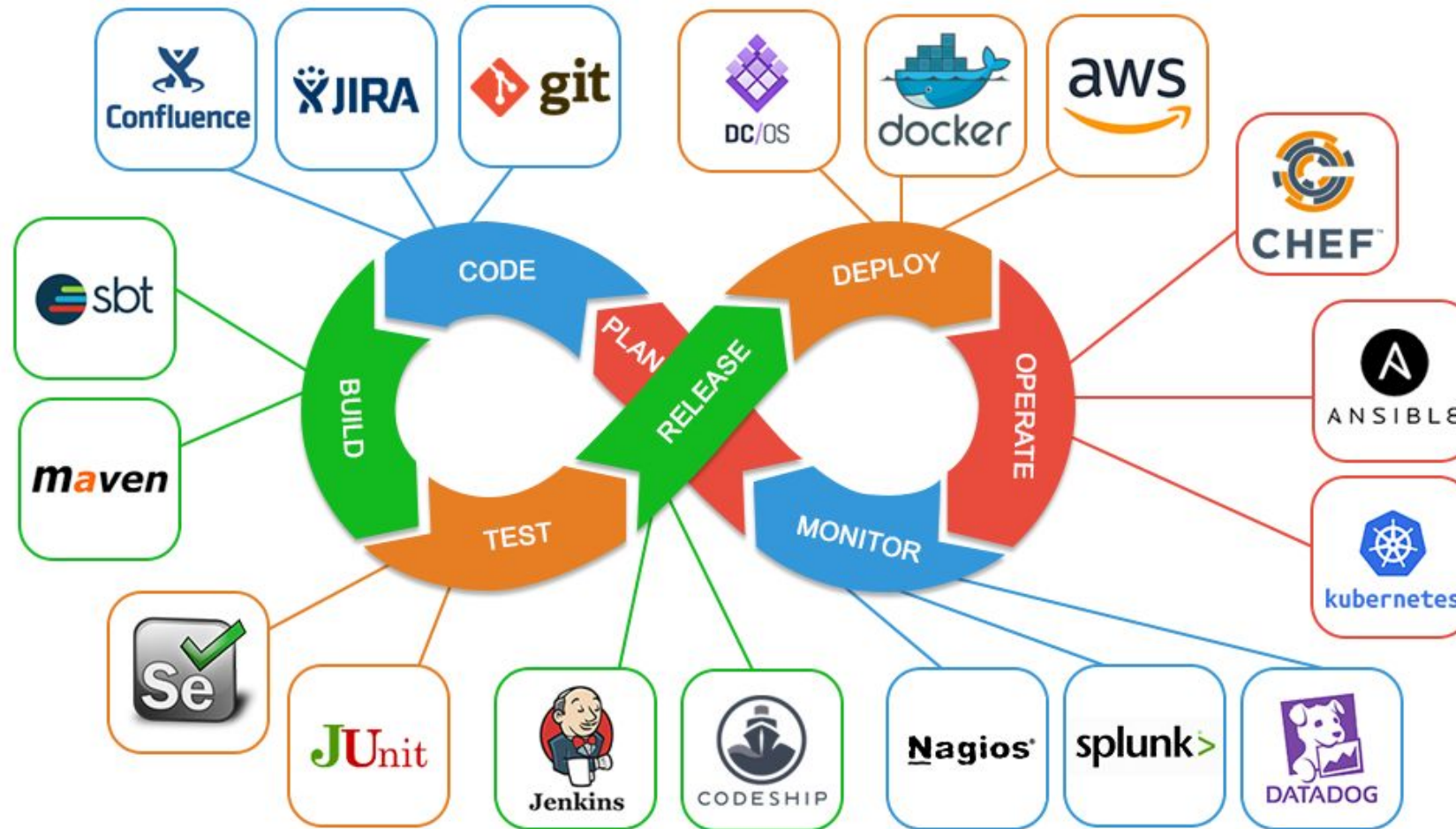


DevSecOps

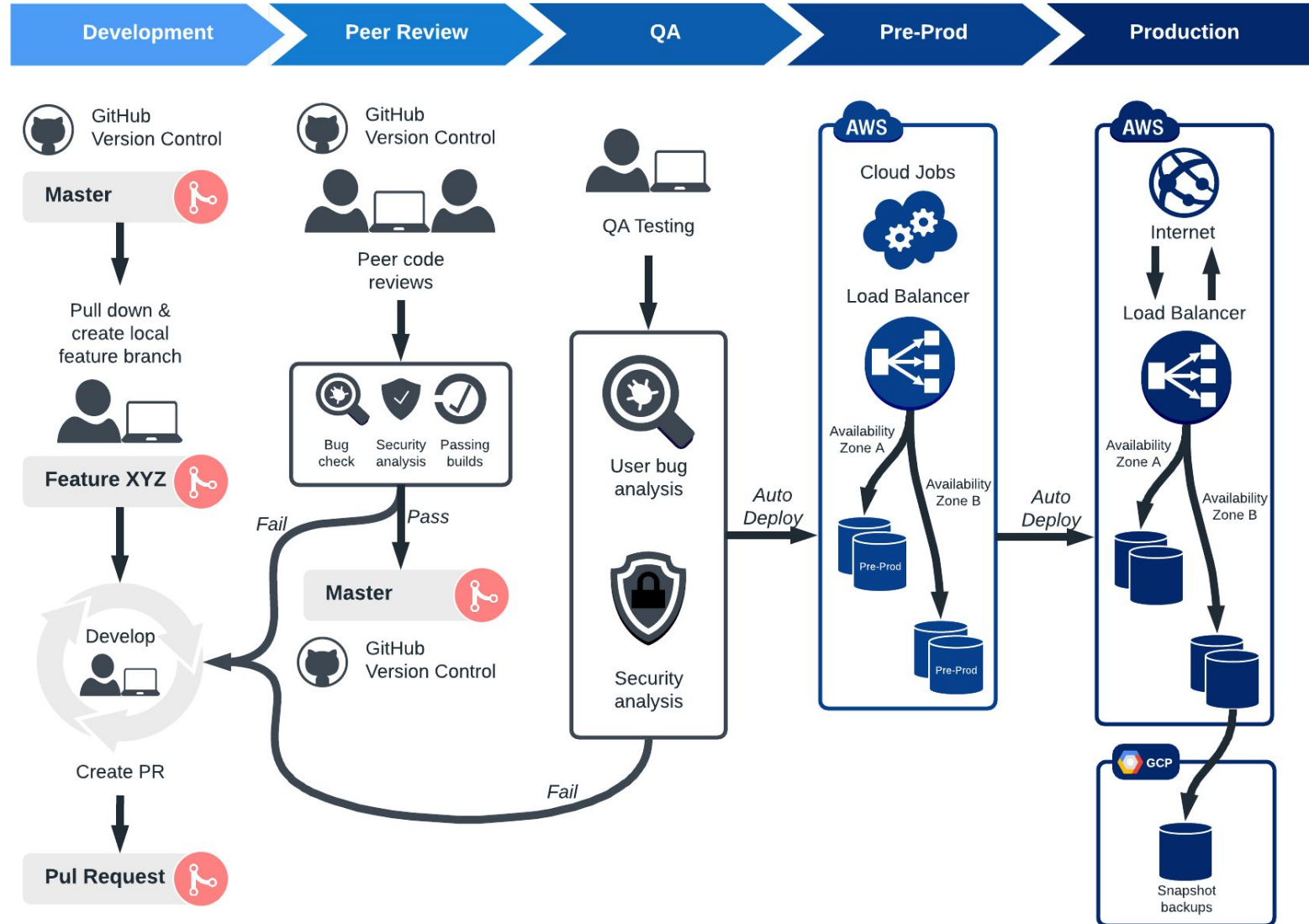




DevOps Tools

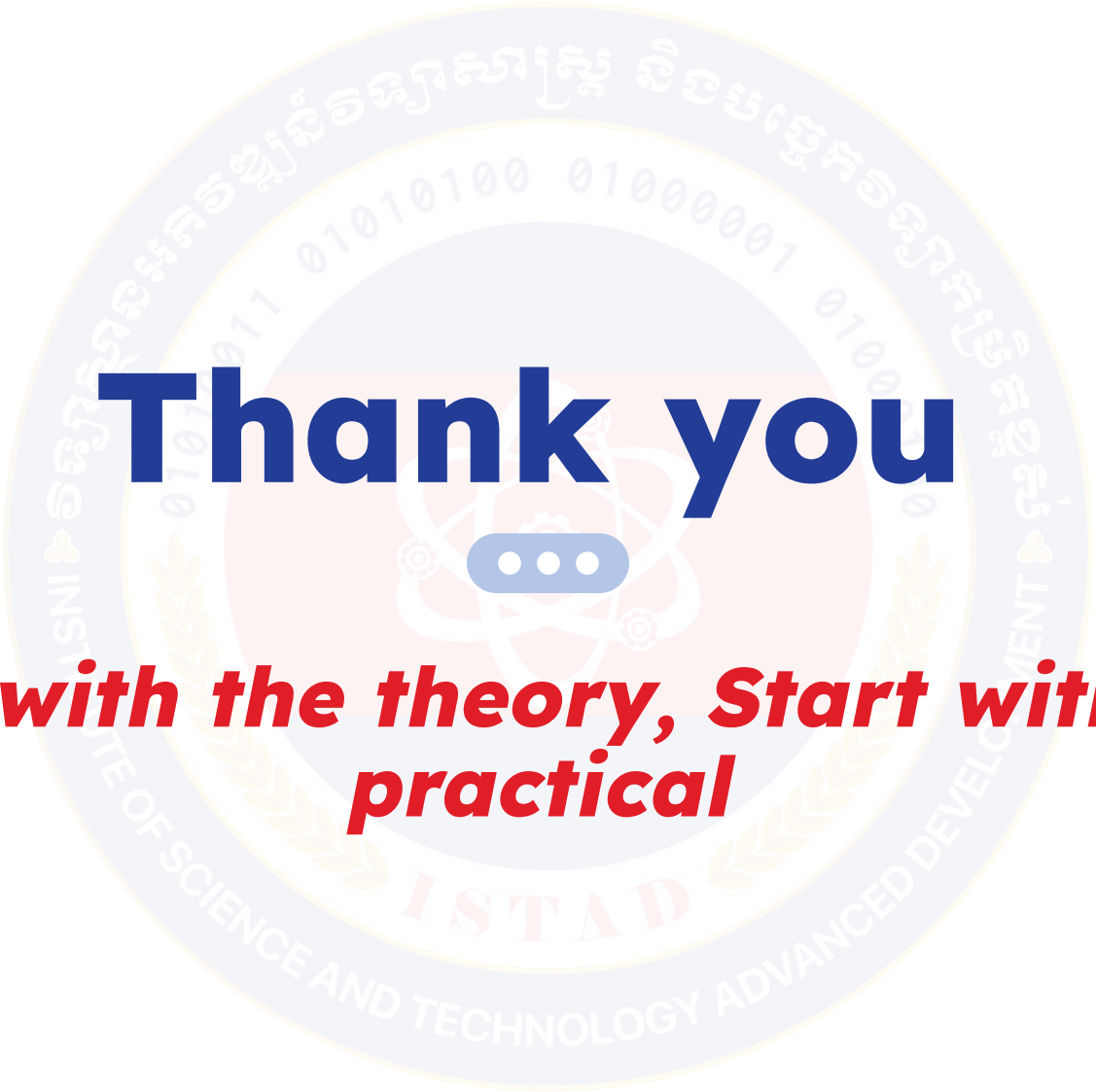


DevOps Tools in Processes



DevOps RoadMap

- Here is the RoadMap for those who want to become DevOps Engineer:
- Link: <https://roadmap.sh/devops>

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Thank you

Begin with the theory, Start with the practical